

Technical Solution Document

Product understanding

GuisedUp presents a feed built around genuine relationships rather than pure engagement. Users authenticate, publish a story, record lightweight interactions, discover semantically related stories, and receive a personalized ranking.

Architecture

Expo React Native app -> Laravel 12 API -> MySQL and Flask embedding service -> SentenceTransformers -> ChromaDB.

Database design

Users have many posts and interactions. Posts belong to one author and have many interactions. Indexed fields support feed and analytics queries.

API design

All feed/search/write endpoints use Sanctum authentication. Login returns a token and logout revokes it. Responses are standardized for mobile consumption.

Feed ranking logic

$0.40 \text{ relationship depth} + 0.30 \text{ semantic similarity} + 0.20 \text{ authenticity score} + 0.10 \text{ recency score}$.

Vector search design

The Flask service uses all-MiniLM-L6-v2 embeddings and stores vectors in ChromaDB. Laravel maps vector IDs back to authorized posts.

Security

Hashed passwords, Sanctum authentication, validation rules, environment secrets, and production TLS recommendations.

Tradeoffs

Application-side ranking and a synchronous vector service were chosen for assessment simplicity and speed.

Future improvements

Queued embeddings, cached feed scores, explicit relationship graphs, moderation, observability, and ML ranking.

AI tools used

OpenAI Codex assisted with scaffolding, implementation, documentation, and testing.